Eliminating Candidates, Not Privacy

Secure Ranked Choice Voting Using MPC

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Voting Systems

Plurality Voting

- Each voter selects one candidate.
- The candidate with the most votes wins.

Advantages

- Simple and easy to understand.
- Quick to count and determine the winner.

Disadvantages

- **Strategic Voting:** Voters may not vote for their true favorite to avoid “wasting” their vote.
- **Limited expressiveness:** Voters can only express a preference for one candidate.
- **Election of minority winners:** The winner might not have majority support.
- **Preceived wasted votes:** Votes for losing candidates do not contribute to the outcome.

Ranked Choice Voting

1. Rank candidates.
2. Check first-choice votes for a majority winner.
3. If no majority, eliminate lowest candidate and redistribute votes.
4. Repeat until a candidate has a majority.

Advantages

- Reduces vote splitting.
- Removes strategic voting (NP-complete). [BTT89]
- More expressive
- Broader support for winners (Potentially).

Disadvantages

- More complex to understand and implement.
- Longer counting process.
- Complex decision-making for voters.

Comparison of systems

Property	Plurality Voting	Ranked Choice Voting
Voter Decision Complexity	Lower	Higher
System Complexity	Lower	Higher
Strategic Voting	Yes	No
Expressiveness	Lower	Higher
Wasted Votes	More	Fewer

Table 1: Summary of properties of Plurality Voting and Ranked Choice Voting

Prior work

“Ranked Choice Voting” using Borda Count

- A secure verifiable ranked choice online voting system based on homomorphic encryption
  by Xuechao Yang, Xun Yi, Surya Nepal, Andrei Kelarev, and Han Fengling
- Secure ranked choice online voting system via intel sgx and blockchain
  by Xuechao Yang, Xun Yi, and Andrei Kelarev

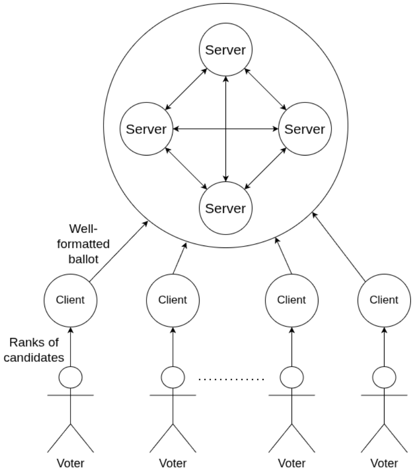
Shortcomings

- Uses Borda Count instead of true Ranked Choice Voting.

Our Contribution

Design and Choices

- **Adversarial Servers:** A subset of servers may be malicious.
- **Honest Voters/Clients:** Voters follow the protocol correctly.
- **Network Communication:** All communication channels are secure and authenticated.



Data Representation

Ballots

- A $C \times C$ matrix B where

$$B[i, j] = \begin{cases} 1 \text{ if candidate } j \text{ is the } i\text{th choice} \\ 0 \text{ otherwise} \end{cases}$$

Example: For 3 candidates and a voter prefered order [2, 0, 1]. The ballot matrix is:

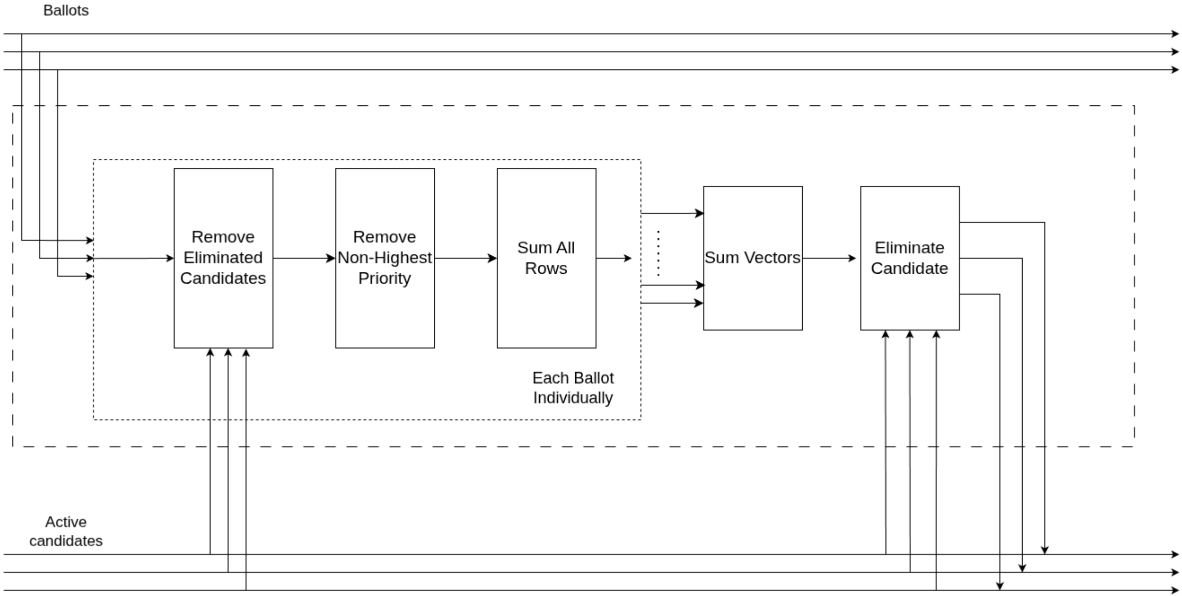
$$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

Active Candidates

- A vector A of length C where

$$A[i] = \begin{cases} 0 \text{ if candidate } i \text{ is eliminated} \\ 1 \text{ otherwise} \end{cases}$$

The Circuit


Ballots
Remove Eliminated Candidates
Remove Non-Highest Priority
Sum All Rows
Each Ballot Individually
Sum Vectors
Eliminate Candidate
Active candidates

Protocol Overview

Remove Eliminated Candidates

$$B_{active}[i,j] = B[i,j] \cdot A[i]$$

Remove Non-highest Candidates

$$B_{highest}[i,j] = B_{active}[i,j] \cdot \left(\prod_{k}^{i-1} \left(1 - \sum_{j} B_{active}[k,j] \right) \right)$$

Sum All Rows

$$v[j] = \sum_{i \in [1,C]} B_{highest}[i,j]$$

Sum Vectors

$$V[j] = \sum_{i=1}^{C} v_i[j]$$

Leakage Model

Minimal Leakage

- Only the winning candidate ID is leaked.
- Candidate to eliminate are computed using secure multiparty computation. (non-trivial gate)

Round based Leakage

- After each round, the full vector of vote counts is leaked. That is V.
- Findind the candidate to eliminate is trivial, and not done using MPC

Experimental Results

(Pretty Pictures)

Varying Number of Servers

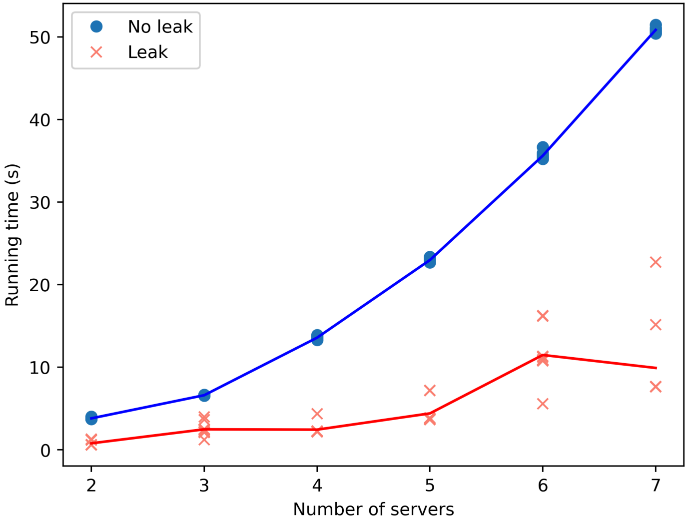


Figure 3: Run time of an election with 3 candidates and 32 voters, varying the number of servers

Varying Number of Voters

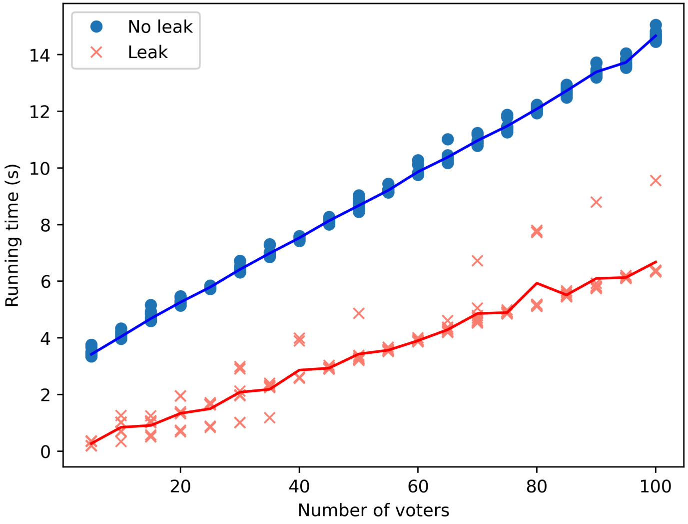


Figure 4: Run time of an election with 3 candidates and varying number of voters, using 3 servers

Varying Number of Candidates

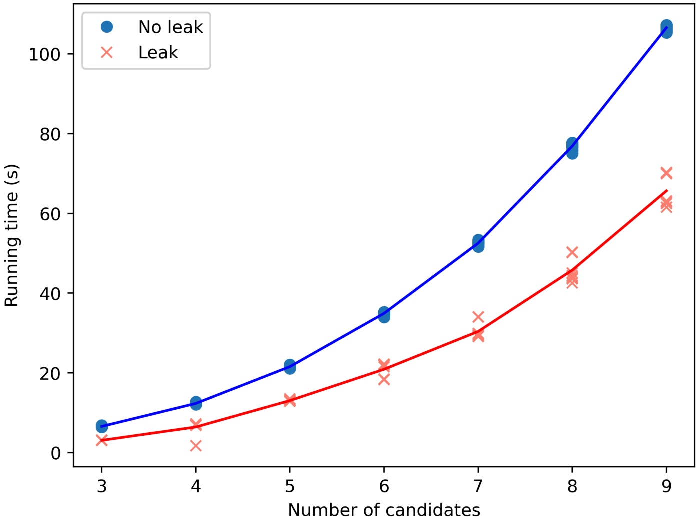


Figure 5: Run time of an election with 32 voters and varying number of candidates, using 3 servers

Candidates scaling

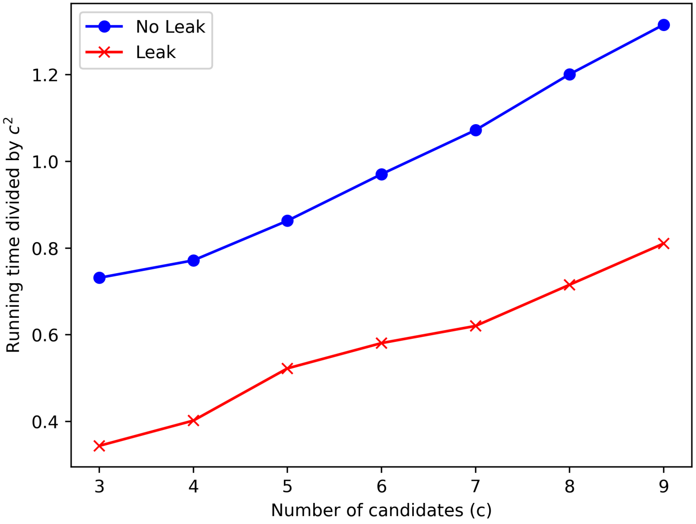


Figure 6: Runtime scaling with number of candidates, showing quadratic

Questions?